INFO5990

**Professional Practice in IT**

# Week 02: Creating organisational value from IT investments

Semester 2, 2026

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*I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.*



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# Agenda

- Organisation
- Differences between value and Organisational value
- Organisational structures

- Role of IT in creating Organisational value
- Aligning IT and Business

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# Introduction



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# What is an Organisation?

- An **Organisation** is a structured group of people working together to achieve specific goals or purposes. It can be formal (like a company, school, government agency, or non-profit) or informal (like a community group or club).
- Key characteristics of an Organisation:
  - **Purpose -** Every Organisation has a mission or objective that guides its activities
  - **People -** It involves individuals with different roles, responsibilities, and skills.
  - **Structure -** There is a defined hierarchy that outlines how tasks are divided and coordinated.
  - **System -** Organisations have established methods for communication, decision-making, and operations.
  - **Resources -** They utilize human, financial, technological, and physical resources.

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# What is the difference between Organisation and Business?

| Organisation | Business |
| --- | --- |
| An **Organisation** is a **broad term** that refers to any group of people working together in a structured way to achieve a common goal. Their purpose is not always profit-driven. | A **business** is a **type of Organisation** whose **primary goal is to make a profit** by providing goods or services. |
| A few examples are:<br>• Non-profits – Red Cross<br>• Governments – NSW Health<br>• Universities – University of Sydney<br>• Clubs – Local sports club | A few examples are:<br>• Startups – Canva<br>• Corporations – Apple<br>• Small Businesses – Local cafes |

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# What is an Organisational Structure?

- An **Organisational structure** defines **how tasks, responsibilities, and authority are distributed** within an Organisation. It shows the **hierarchy**, **reporting lines**, and how different roles and departments **interact and coordinate** with each other.

- Why is the Organisational structure an important topic to understand?
  - It clarifies who reports to whom
  - Defines decision-making authority
  - Improves efficiency and communication
  - Helps manage growth and change

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# Types of Organisational structures

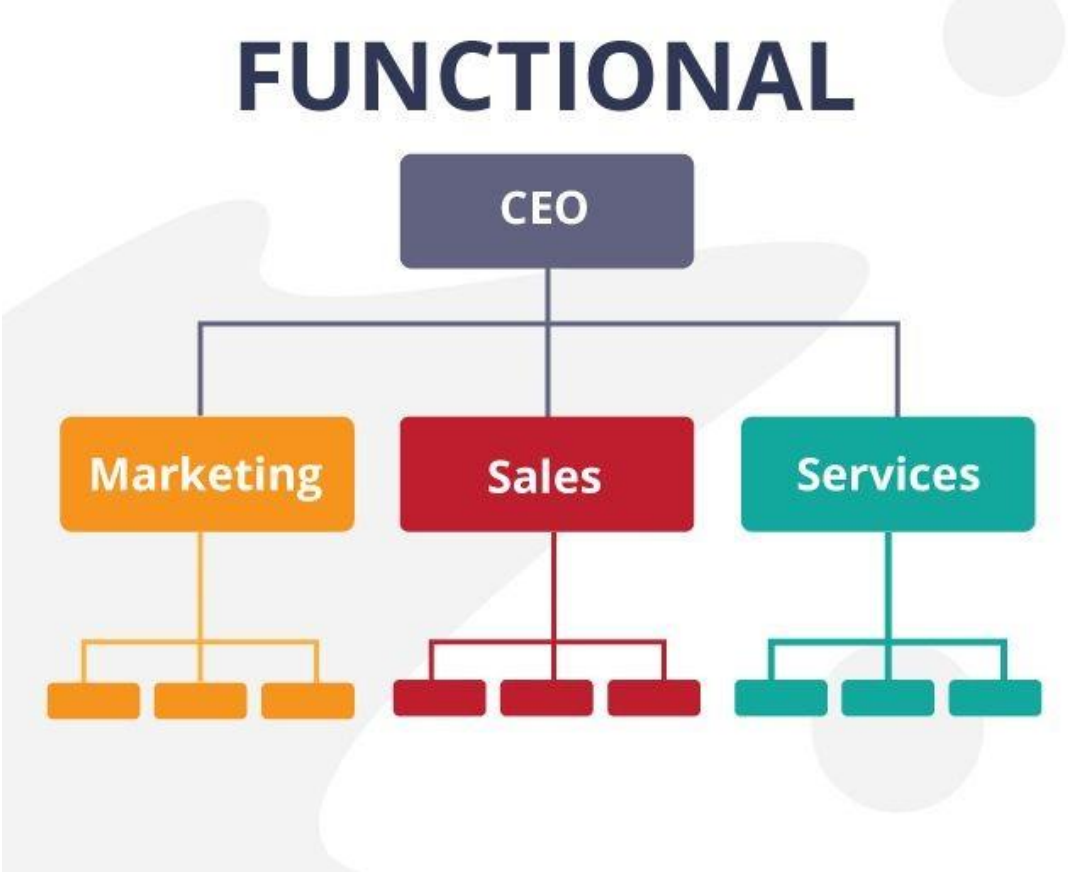
- Functional
  - Groups employees based on specialized roles or functions.
- Matrix
  - Employees report to two managers - one for function, one for project.

- Flat
  - Few or no levels of middle management. Common in startups or small Organisations.
- Hierarchical
  - Traditional structure with clear top-down command and multiple layers of authority.

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# Types of Organisational structures - Functional

- An organisation is divided into departments based on specific functions (e.g., IT, HR, Finance, Marketing).
- Pros:
  - Specialisation leads to efficiency
  - Clear roles and responsibilities
  - Easier to manage and scale by department
- Cons:
  - Limited communication across departments
  - Can create silos and reduce flexibility
- Best for:
  - Large or growing organisations with clearly defined departments (e.g., tech companies, banks).




(Ivanius, 2023)

# Example #1 – Functional Organisational structures

- Microsoft was organised functionally, with departments like Engineering, Sales, Marketing, HR, and Finance operating independently. Each function had its own VP reporting to the CEO.

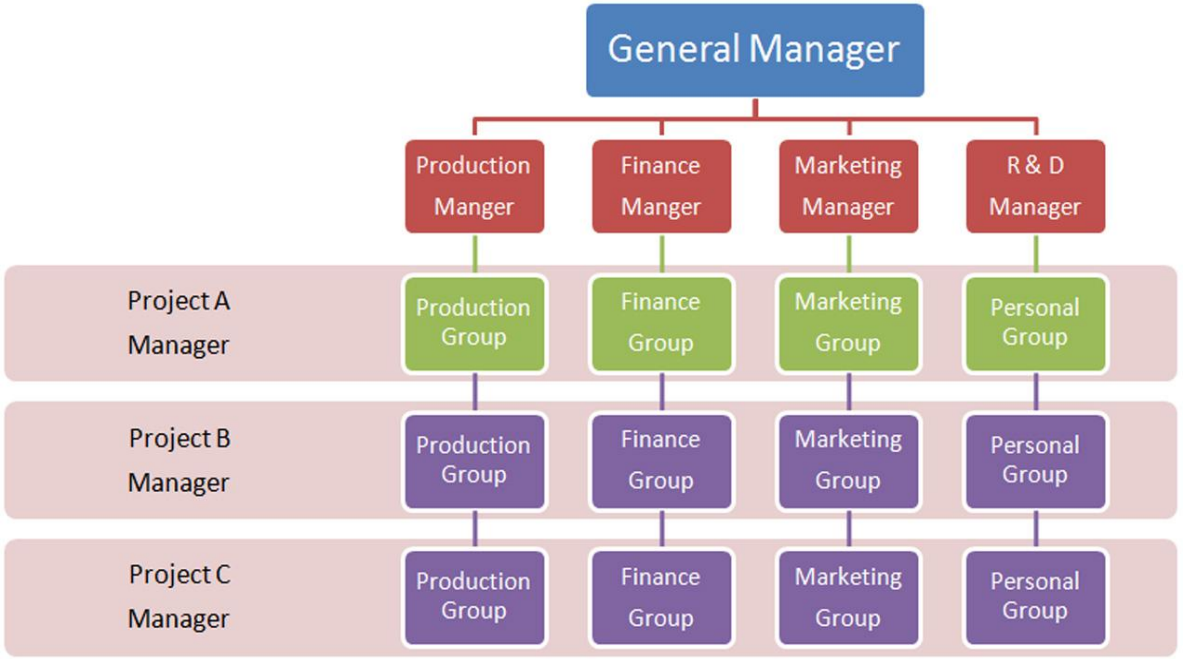
- Why did it work?
  - Specialisation in each domain helped the company scale globally while maintaining departmental focus.




(Microsoft, 2012)

# Types of Organisational structures - Matrix

- Employees report to more than one manager, typically by function and project/team.
- Pros:
  - Efficient use of resources across projects
  - Encourages teamwork and cross-functional skills
  - Increases flexibility and adaptability
- Cons:
  - Dual authority can cause confusion
  - Requires strong communication and coordination
- Best for:
  - Project-based or global companies (e.g., consulting firms, R&D, IT services companies).




(Anon, nd)

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# Example #2 – Matrix Organisational structures

- Employees at Google often report to both a product manager (e.g., for Android or YouTube) and a functional manager (e.g., Engineering or UX).

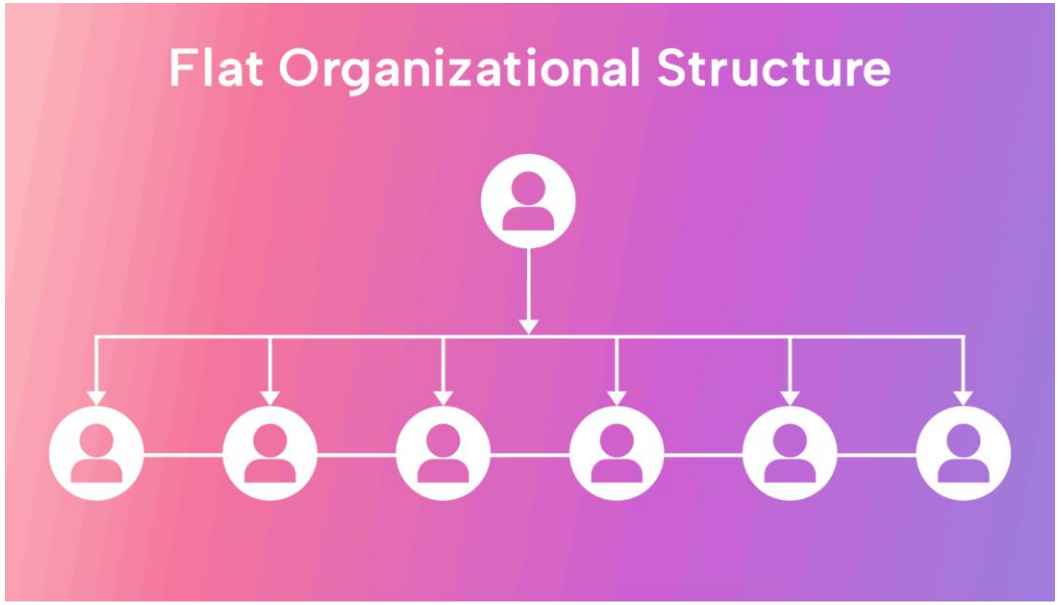
- Why it worked?
  - This dual-reporting allows for better collaboration across projects while maintaining technical excellence and innovation.




(Google, 2015)

# Types of Organisational structures - Flat

- Few or no levels of middle management between staff and leadership. Everyone is closer to decision-making.
- Pros:
  - Fast communication and decision-making
  - Encourages innovation and collaboration
  - Flexible and adaptive
- Cons:
  - Can lead to confusion in roles/responsibilities
  - Not scalable for large teams
- Best for:
  - Startups or small teams where agility and speed matter more than hierarchy.




(Motion, 2023)

# Example #3 – Flat Organisational structures

- Valve corporation famously operates with *no formal hierarchy*. Employees choose what projects to work on, and everyone contributes equally in decision-making.

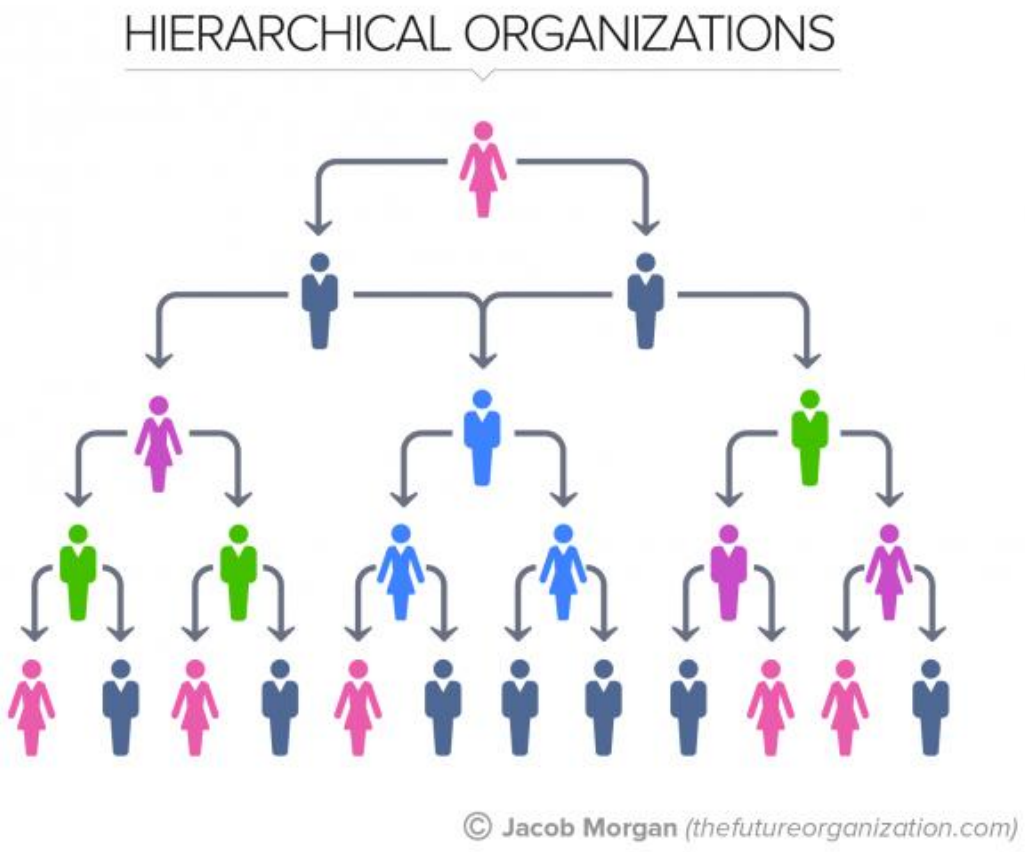
- Why it worked?
  - For a creative and agile industry like gaming, this structure fostered innovation and rapid iteration.




(Valve, 2018)

# Types of Organisational structures - Hierarchical

- Traditional top-down approach with multiple levels of authority and a clear chain of command.
- Pros:
  - Clear reporting lines and accountability
  - Stable and easy to manage large teams
  - Supports growth and delegation
- Cons:
  - Slower decision-making
  - Can stifle creativity and employee autonomy
- Best for:
  - Government bodies, large enterprises, and traditional corporations.




(Morgan, 2015)

# Example #4 – Hierarchical Organisational structures

- IBM follows a classic hierarchical structure with multiple layers of management. Each department has a clear chain of command from junior employees up to executives.

- Why it worked?
  - In such a massive enterprise, a strict hierarchy ensures stability, control, and accountability across global operations.



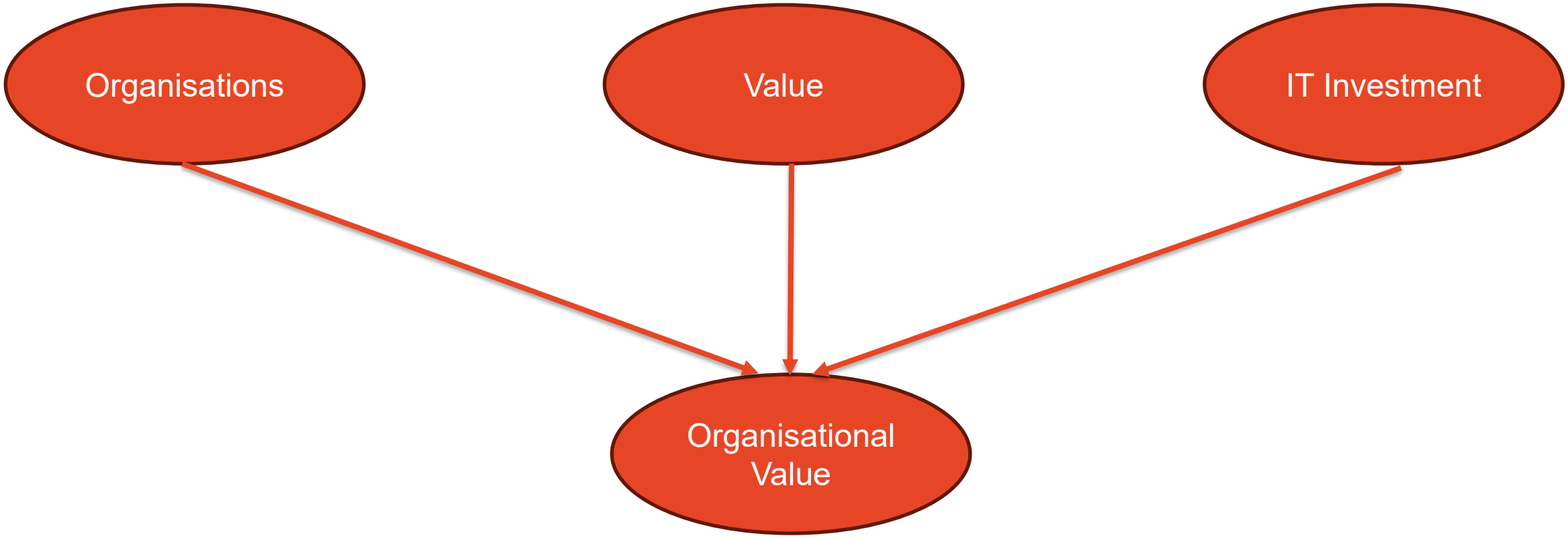
(IBM, 2025)

# Role of IT in creating Organisational Value



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# Four Foundational concepts




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# Value and Organisational value

- Value
  - **Value** refers to the **worth**, **usefulness**, or **benefit** of something. It can be:
    - Tangible (Example – Money, Products, etc.)
    - Intangible (Example – Trust, Satisfaction, etc.)
  - In general value answers: *"How does this benefit me or others?"*

- Organisational Value
  - **Organisational value** refers to the **value created, delivered, and sustained by an Organisation** for its **stakeholders,** including customers, employees, shareholders, and society.
  - In general, Organisational value answers: *"How does the Organisation create meaningful outcomes for its stakeholders?"*

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# What is an IT investment?

- An IT investment refers to the allocation of financial, human, and technological resources into information technology systems, tools, or services to support or improve an Organisation's operations, performance, or strategic goals.

- These resources and capabilities can be internally owned by the Organisation (insourced) or externally sourced (outsourced)

- Examples:
  - Hardware: Individual devices, servers, networks, etc.
  - Software: Apps, platforms, middleware, etc.
  - Data, People, Processes and Frameworks, etc.

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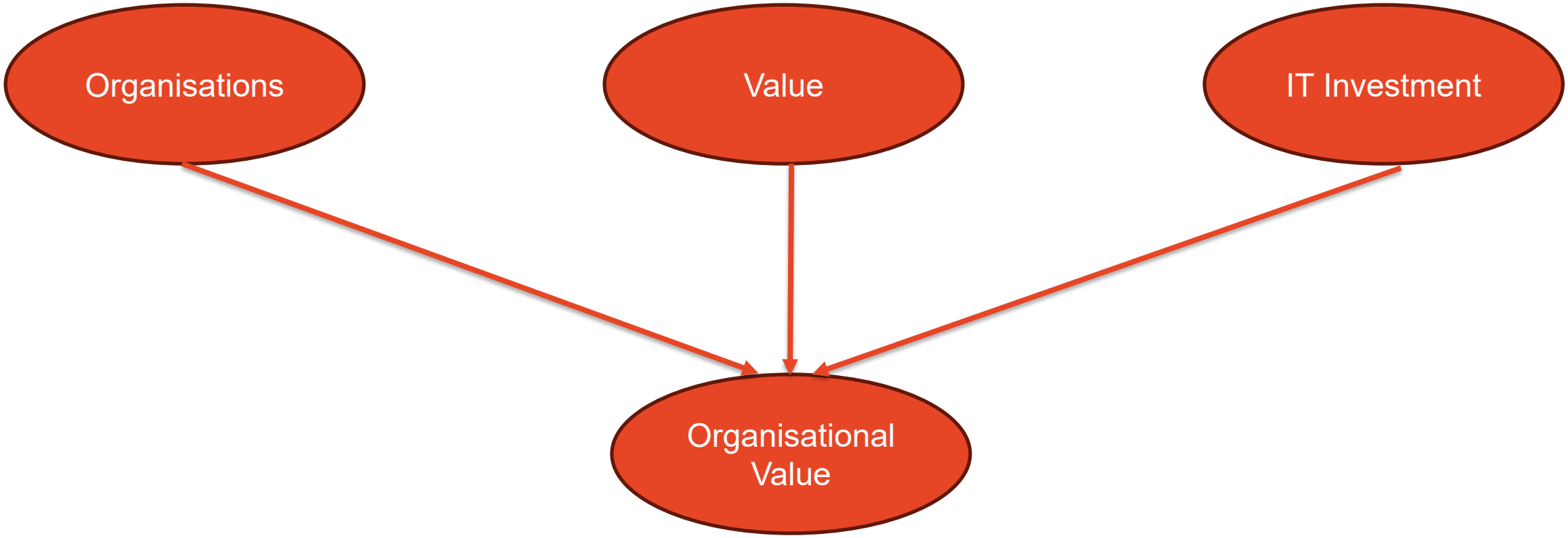
# Distinction between resources and capabilities

- Resources refer to what an Organisation owns. They might be:
  - Tangible: Such as property, machinery, hardware, etc.
  - Intangible: Such as knowledge, skills of employees, policies, etc.

- Capabilities refer to what an Organisation can do by using those resources.
  - A capability is the ability to execute a specified course of action to achieve certain outcomes based on the resources available such as skills and technology

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# So how do these concepts link together?




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# So how do these concepts link together?

- Organisation
  - Structure, people, and processes that drive operations.
  - Provides the foundation for strategy and delivery.
- IT Investment
  - Technology enables efficiency, innovation, and scale.
  - Examples: Automation, data analytics, cloud, AI.
- Value
  - The result of organisational effort and tech enablement.
  - Benefits for stakeholders: customers, employees, investors, society.
  - Examples: Profits, Dividends, Operational efficiency, etc.

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# The critical role of IT in creating Organisational Value

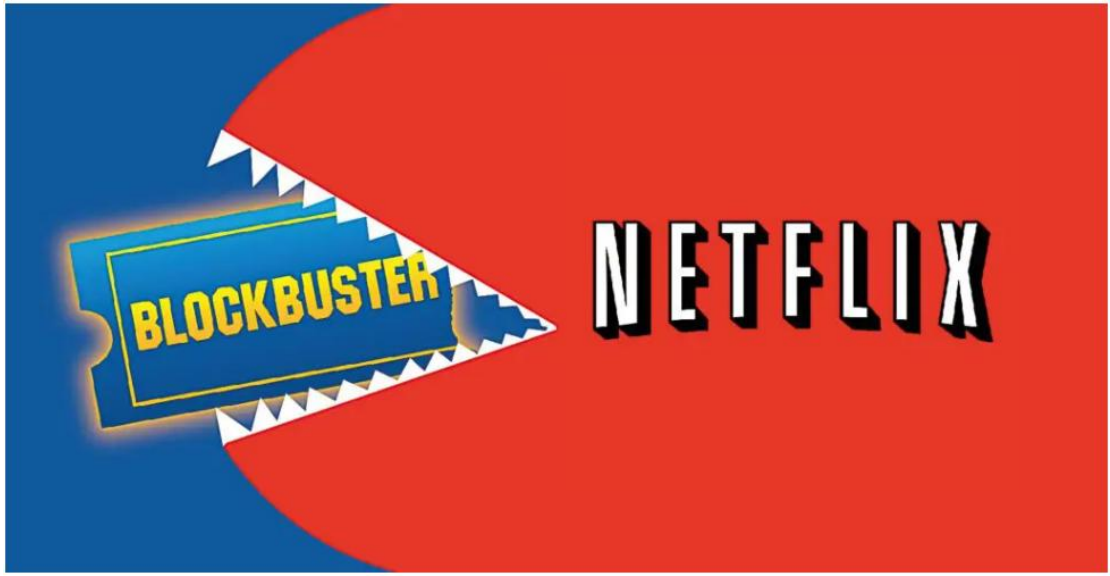
- Information Technology (IT) plays a central role in helping Organisations create, deliver, and sustain value.
- In today's digital world, IT is not just a support function - it's a strategic enabler of innovation, efficiency, and competitiveness.

- But how?
  - Improving operational efficiency – automation, cloud systems, databases
  - Enhances decision making – Data analytics, BI tools, Artificial Intelligence
  - Drives innovation – IoT, Blockchain, Agentic AI
  - Enables customer-centric strategies – CRM systems, Self-service platforms
  - Supports scalability and flexibility – AWS, GCP, Azure

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# Case study #1 – IT investments increase Organisational value

- Netflix invested in:
  - Streaming Infrastructure: Enabled global, 24/7 content access.
  - Recommendation Algorithms: Increased watch time and user satisfaction.
  - Cloud Computing (AWS): Scalability without hardware limitations.
  - Content Analytics: Informed original content creation (e.g., House of Cards).
- Organisational value outcomes:
  - Customer-Centric Model: Personalized experiences and binge-worthy content.
  - Rapid Scalability: 200M+ global subscribers.
  - Disrupted the Market: Became a leader in digital entertainment.
  - Sustainable Competitive Advantage: Data-driven content decisions, operational efficiency.




Blockbuster: (Francis, nd.)

- Founded in 1985
- Failed to adapt to the shift toward online streaming and digital rentals.
- Late and poorly executed attempts to launch a streaming service and impose unpopular fees (like late fees) hurt customer loyalty.

# A quick 5 min stretch

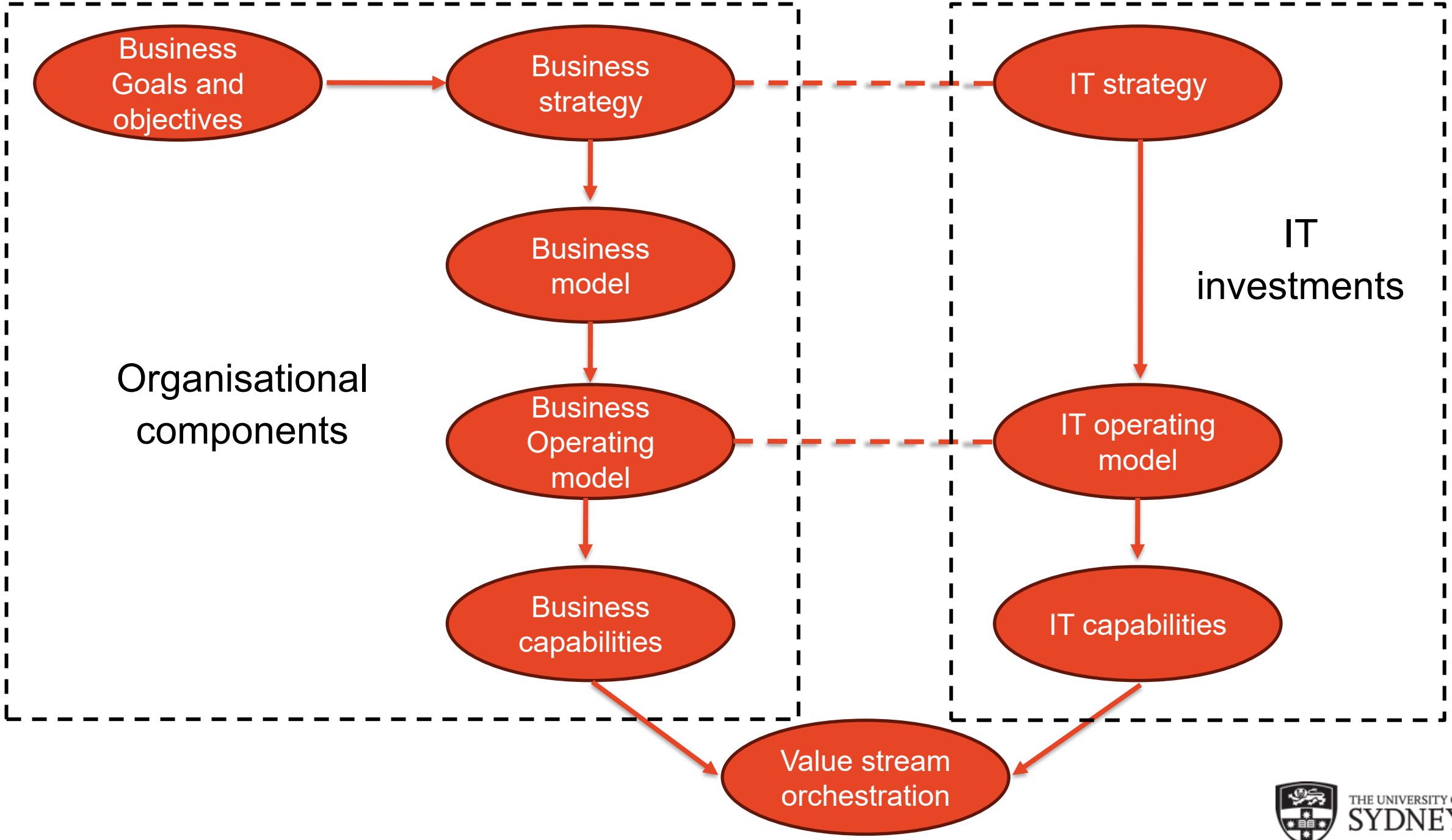


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# Aligning IT and Business



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# The 2 sides

- Organisation side
  - This side outlines how an Organisation builds value through **business planning and operational structure**:
  - Business goals and objectives, business strategy, business models, business operating models, and business capabilities.

- IT investment side
  - This side shows how **IT must align with business** to support and enable value creation:
  - IT strategy, IT operating model, and IT capabilities.

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# Business goals and objectives

- Business goals and objectives are the foundation of all Organisational activity.
- They define the **vision**, **mission**, and **desired outcomes**.
- These guide both business and IT strategies.

**Value begins with a clear understanding of what the Organisation wants to achieve.**

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# Business goals and objectives contd.

- A business goal defines what the organisation ultimately wants to achieve**.** It provides direction and purpose.
  - **Example:** Increase customer satisfaction in the next 12 months.
- Objectives specify measurable targets and outcomes that help an organisation achieve its goals.
  - **Examples**: reduce expenses by 5%, increase sales by 5% etc.. Reduce defects by 50% reduce incidence of a childhood disease by 50%
- Business goals vs Business objectives
  - Business goals represent the **direction** in which a company intends to go and define what the Organisation wants to achieve.
  - A business objective specifies the **methods** and paths that can help a business achieve that goal.

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# Example of aligning Business and IT

Let’s consider a startup called "QuickCart", an on-demand grocery delivery platform, and map it to the categories in the diagram.

- Business Goal and Objective of "QuickCart"
  - **Business goal:** Become a preferred ultra-fast grocery-delivery service for customers in urban areas.
    This describes the broad, long-term outcome the business wants to achieve.
  - **Business objective:** Complete at least **90% of grocery deliveries within 30 minutes by the end of the next 12 months**.

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# Business strategy

*The High-level plan*

- Business strategy outlines **how the Organisation will achieve its goals**.
- This includes choices around target markets, competitive positioning, resource allocations, etc.
- IT must be aligned with this strategy to be effective.
- Strategy drives both structure and technology direction.

## Business Strategy of QuickCart

- **Business Strategy:** Differentiate through **fast delivery**, **real-time order tracking**, and **AI-driven logistics**.
- **Competitive Advantage:** Deliver groceries **faster and more efficiently** than traditional supermarkets.

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# IT strategy

***Technology vision supporting business goals***

- IT strategy is the roadmap for **how technology will support business strategy**.
- It focuses on digital transformation, infrastructure decisions, innovation plans, etc.
- *Key Point:* IT strategy must be developed in lockstep with business strategy.

## IT Strategy of QuickCart

- **Cloud-first infrastructure** to ensure scalability and high availability.
- **AI-driven logistics** for route optimisation and delivery prediction.
- **Real-time tracking system** for customers and delivery drivers.
- **Mobile app and digital payment platform** for a seamless customer experience.
- **Data analytics** to forecast demand and optimise inventory.

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# Business models

*Translating strategy into action*

- The business model defines **how value is created, delivered, and captured**.
- It can be best show as a Business Model Canvas (BMC)
- Business models guide IT system requirements (e.g., eCommerce platforms, CRMs).
- A solid model ensures the strategy is executable.

## Business Models of QuickCart

- QuickCart uses a **digital platform model** connecting customers, grocery stores and delivery partners.

**Revenue streams:**

  - Delivery fees
  - QuickCart Prime subscriptions
  - Commissions from partner stores

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# The Business Model Canvas

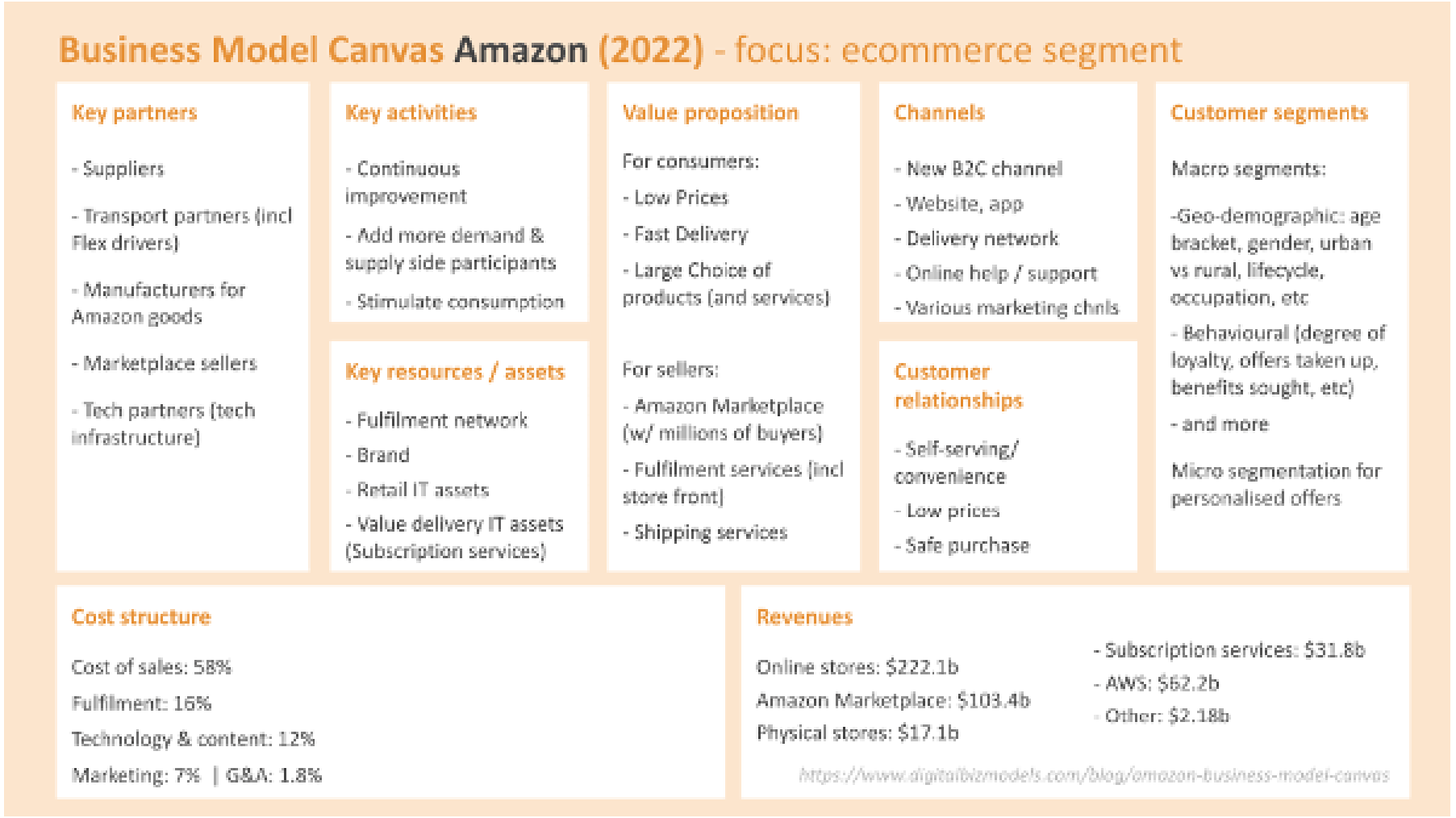
| Designed for: | Designed by: | Date: | Version: |
|---|---|---|---|
| | | | |

| Key Partnerships | Key Activities | Value Propositions | Customer Relationships | Customer Segments |
|---|---|---|---|---|
| | | | | |
| | **Key Resources** | | **Channels** | |

| Cost Structure | Revenue Streams |
|---|---|
| | |

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# An example of the BMC




(DigitalBizModels, 2022)

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# Business operating model

*How the business functions*

- The business operating model defines **how the organisation is structured and operates**.
- It describes centralization vs decentralization, business units and interdependencies, decision making authority, etc.
- Determines the IT support and governance model required.
- Operating model alignment enables efficient IT integration.

## Business Operating Model of QuickCart

QuickCart uses a **platform-based, partner-enabled operating model**:

- Partner stores prepare customer orders.
- Independent delivery partners complete deliveries.
- QuickCart centrally manages the digital platform, payments, order allocation and customer experience.

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# IT operating model

***How IT is organized and delivered***

- Defines **how IT functions are structured and governed**.
- This includes IT service delivery, vendor management, project management, agile practices, etc.
- Must align with the business operating model.
- Efficient IT operations are critical for smooth execution.

## IT Operating Model QuickCart

- A central IT team manages the digital platform, data and cybersecurity.
- Cross-functional Agile teams develop and improve customer-facing services.
- Cloud infrastructure supports scalability and availability.
- DevOps practices enable frequent and reliable software releases.

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# Business capabilities

***What the business can do!***

- Capabilities are the **skills, processes, and resources** the business uses to deliver value.
- Examples: logistics, R&D, customer service, data analytics.
- Must be supported and often enhanced by IT (e.g., automation, AI, CRM systems).
- IT enables, enhances, or transforms these core capabilities.

## Business Capabilities of QuickCart

- **Demand forecasting** – anticipating customer demand and maintaining product availability.
- **Delivery and route optimisation** – planning efficient routes and reducing delivery times.
- **Order processing and fulfilment** – receiving, assigning and completing customer orders.
- **Payment processing** – securely accepting digital wallets, UPI and credit-card payments.

# Key components – IT capabilities

*What IT can deliver*

- IT capabilities refer to the **technical strengths** of the IT function.
- For example, system integration, cloud management, data analytics and reporting, cybersecurity, etc.
- These directly support and scale business capabilities.
- Strong IT capabilities are essential for digital competitiveness.

- **IT Capabilities (Enabling Execution) of QuickCart**
  - **Cloud infrastructure management** – supports scalable and reliable operations.
  - **AI-enabled customer support** – provides fast responses to customer queries.
  - **Real-time tracking integration** – enables customers to track deliveries.
  - **Data analytics capability** – supports demand forecasting and operational decisions.
  - **Cybersecurity and payment security** – protects customer and transaction data.

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# The integration point – Value stream orchestration

***Where Business and IT Truly Meet***

- Value stream orchestration is about **coordinating business and IT to deliver continuous value**.
- It involves synchronizing people, processes, and technology to **maximize the value delivered to stakeholders**.
- Enables smooth flow of value from idea to delivery.
- Orchestration ensures technology investments produce real business outcomes.

**Value Stream Orchestration (Bridging Business & IT) QuickCart**

Aligning IT & Business Goals by ensuring AI, cloud, and automation work together to:

- Enhance delivery speed.
- Increase efficiency of logistics partners.
- Improve customer satisfaction & retention.

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# Outcome? - Organisational value

***The Final Outcome***

When **business and IT** are properly aligned and orchestrated, the result is **Organisational value.**

- The benefits are measurable such as:
  - Increased customer satisfaction
  - Operational efficiency
  - Revenue growth
  - Innovation
  - Competitive advantage

## Organizational Value (Core Values) QuickCart

- Customer Obsession (Prioritizing speed & reliability in every order)
- Innovation (Continuously evolving delivery models)

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# Best practices for IT investments

- Ensure every IT investment supports overall business goals.
- Engage business leaders in IT planning.
- Set measurable outcomes for each IT investment (e.g., cost savings, process efficiency, customer experience).
- Use KPIs to track ROI over time.
- Use a cost-benefit or value-risk matrix to rank IT initiatives.
- Focus on projects that provide both short-term wins and long-term value.
- Invest in technologies that can grow with the business (e.g., cloud, APIs, modular platforms).
- Avoid systems that lock the business into rigid structures.
- Encourage experimentation with emerging tech (AI, blockchain, IoT) in low-risk areas.
- Balance innovation with practical business value.

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# For you!



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# Example of aligning Business and IT

Let’s consider a startup called "QuickCart", an on-demand grocery delivery platform, and map it to the categories in the diagram.

- Business Goal and Objective
  - Business Goal: Deliver ultra-fast grocery delivery services to urban customers. This is the high-level purpose of the business, what it ultimately aims to achieve.
  - Business Objective: Provide grocery delivery within 30 minutes. This is the specific, measurable action that supports the goal.
- Business Strategy
  - Strategy: Differentiate through speed and real-time tracking, supported by AI-driven logistics.
  - Competitive Advantage: Faster than traditional supermarket deliveries.
- IT Strategy
  - Tech-First Approach (AI, cloud computing)

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# Example of aligning Business and IT

- Business Models
  - Revenue Model:
  - Delivery Fees (charged per order).
  - Subscription Model (QuickCart Prime for free deliveries).
  - Partnered Store Commissions (grocery stores pay a fee per sale).
- Business Operating Model
  - Customers order via an app or website.
  - AI assigns the nearest delivery partner (crowdsourced drivers).
  - Local partner stores fulfill the orders within minutes.
  - Real-time tracking enables customers to monitor deliveries.
- IT Operating Model
  - Mobile-App design for quick app-based ordering.
  - Machine learning for predicting demand and stock-outs.
  - Data analytics to optimize driver routes.

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# Example of aligning Business and IT

- Business Capabilities
  - AI-Driven Demand Forecasting ( Ensures stock availability at partner stores.)
  - Fleet & Route Optimization (AI minimizes delivery time through smart navigation.)
  - Automated Order Processing (Orders are auto-assigned to the fastest driver.)
  - Integrated Payment System (Seamless checkout with wallets, UPI, and credit cards.)
- IT Capabilities (Enabling Execution)
  - Cloud-based platform
  - AI-powered chatbot ( Handles customer queries instantly)
  - GPS integration (Real-time delivery tracking for customers)
- Value Stream Orchestration (Bridging Business & IT)
  - Aligning IT & Business Goals → Ensuring AI, cloud, and automation work together to:
  - Enhance delivery speed.
  - Increase efficiency of logistics partners.
  - Improve customer satisfaction & retention.

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# Example of aligning Business and IT

- Organizational Value (Core Values)
  - Customer Obsession (Prioritizing speed & reliability in every order)
  - Innovation (Continuously evolving delivery models)

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# End of lecture questions

- Define the term “Organisation” and explain how it differs from a “business”.
- What is the difference between value and organisational value? Provide an example to illustrate.
- Explain the distinction between resources and capabilities with examples.
- List and briefly describe any three types of organisational structures.
- What is ‘value stream orchestration’ in the context of IT and business alignment?

# Case-study based questions

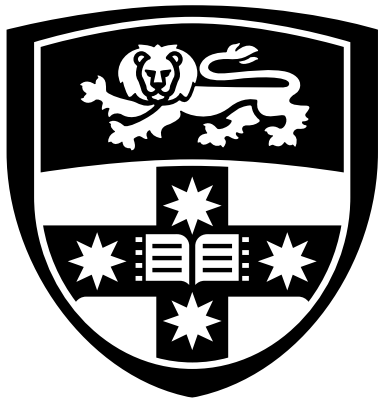
- BrightCare is a mid-sized private healthcare organisation that specializes in telehealth services and digital patient records. The management has identified a need to increase organisational value by improving service delivery, enhancing patient satisfaction, and reducing operational costs. Currently, their systems are outdated, departments are siloed, and reporting is mostly manual.
- The company has decided to:
  - Invest in a new cloud-based health information system (HIS)
  - Hire a Chief Information Officer (CIO) to align IT strategy with the company's vision
  - Restructure from a functional model to a matrix structure to enable cross-departmental collaboration
- The CIO proposes the following plan:
  - Define clear business goals (e.g., 30% reduction in wait times, 95% digital record accuracy)
  - Align IT and business strategies by upgrading technology and retraining staff
  - Build capabilities in data analytics and real-time reporting
  - Implement value stream orchestration to connect IT workflows with patient care delivery

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# Case-study based questions

- Identify two business goals BrightCare is trying to achieve through IT investment. How do these relate to Organisational value?
- What kind of Organisational structure was initially in place at BrightCare, and why is the move to a matrix structure beneficial in this context?
- Explain the role of the CIO in ensuring alignment between IT and business at BrightCare. What challenges might the CIO face?
- Evaluate whether the planned IT investment is likely to create organisational value. Justify your answer using concepts from the lecture.

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